An even attaching group and the critical symmetric wing

Human Compression project: a component of the L019 recovery

An endpoint anchor of an ordered involution family is a point moved by its first member and fixed by every later member. All groups below are actual permutation groups. An attachment from outside its domain at an anchor a has the form $(a\ c)\gamma$, where c is new and γ is an involution or identity supported off the old domain and c. In particular, γ centralizes every old mark and the pure transposition $(a\ c)$.

Lemma 1. *Let $(t_0, \dots, t_4)$ be a natural rank-five C-marking of A_{10}. Suppose a is an endpoint anchor and its tail is transitive on the other nine points. No attachment $(a\ c)\gamma$ as above can precede this marking in a string C-marking.*

Proof. We acquire the complete ordered rank-five list from the accepted small-degree part of the alternating rank theorem (L044). Its retained terminal list has ten rank-five tuples: eight generate A_{10}, while the other two generate groups of order 1920. We retain and scan this complete superset. The completeness of its A_{10} portion, including the original census validation, is an acquired classification premise; the following operation only filters that list.

For each retained tuple in each orientation, inspect every point for the anchor condition, then inspect the tail orbits. Exactly four oriented records pass, all belonging to the eight A_{10} tuples. The table uses zero-based source indices and orientation 0 for the recorded order, 1 for reversal. Set

$$u = (a\ 11), \quad U = \langle t_0, t_1, t_2, t_3 \rangle, \quad V = \langle u, t_0, t_1, t_2 \rangle, \quad M = \langle t_0, t_1, t_2 \rangle.$$

Index	Orientation	Anchor	$\lvert U\rvert$	$\lvert V\rvert$	$\lvert U \cap V\rvert$	$\lvert M\rvert$
1	0	10	5040	5040	360	60
2	1	1	5040	5040	360	60
4	1	1	40320	10080	720	120
7	0	10	40320	10080	720	120

The certificate acquires all ten literal tuples from the retained list, rather than selecting only these four examples. The separate GAP check identifies their exact group orders as well. A direct permutation implementation enumerates each of the displayed finite closures and intersects their elements; a separate GAP calculation agrees. Every row gives $[U \cap V : M] = 6$.

For an actual attachment $u\gamma$, the corresponding group on the five entries $u\gamma, t_0, t_1, t_2, t_3$ lies in the direct product of the pure group and $\langle \gamma \rangle$. Projection onto the pure group has central kernel of order at most two. The acquired central intersection lemma says that, if the actual marking had IP, the index $[U \cap V : M]$ would divide a power of two: its difference quotient embeds in a quotient of that central kernel. The factor three in six contradicts this necessary condition. This includes $\gamma = 1$ and proves the assertion. □

Theorem 2. *Let $(t_0, \dots, t_{k-1})$ be a faithful primitive string C-marking on a set Y of size $d \geq 3$, consisting entirely of even permutations. Suppose it has an endpoint anchor a, its tail is transitive on $Y \setminus \{a\}$, and an attachment from outside at a occurs immediately before it in a larger string C-marking. Then either*

$$2k \leq d - 1, \tag{1}$$

or $d = 6$, $k = 3$, the group is the primitive degree-six copy of A_5, and its tail has order ten.

Proof. The group is not S_d, because every mark is even. First suppose it is the natural A_d. The complete acquired alternating rank bound gives (1) for $d \geq 12$. Its exact small-degree conclusions leave only A_5 of rank three, A_{10} of rank five, and A_{11} of rank six as possible violations of (1).

The degree-five case has no endpoint anchor. Indeed a nonidentity involution of A_5 has one fixed point. The third mark fixes a, so its unique fixed point is a; the first mark commutes with it and therefore fixes a, a contradiction. The degree-ten case is excluded by Lemma 1, using the explicit tail-transitivity hypothesis. For degree eleven, the acquired actual-wing certificate inspects all four complete rank-six tuples and both orientations. Each nonempty anchor gives the same pure-attachment prefix index six, with orders $5040, 5040, 360, 60$. The same central-kernel argument excludes any attachment $(a\ c)\gamma$. Here the attachment is already pure on $Y \cup \{c\}$, so no separate centralizer normalization or general quotient-IP assertion is needed.

For every other primitive group, the complete primitive theorem acquired in L047 applies to the actual faithful C-marking. Unless its rank is strictly less than $d/2$, it belongs to the twelve retained ordered small representatives: degree-six copies of A_5 or S_5, or the degree-ten rank-five copy of S_6. The already acquired twenty-four-orientation filter records every anchor and complementary tail orbit. Without imposing its additional character test, the even groups with a transitive tail are exactly source indices and orientations

$$(0,1),\ (1,0),\ (1,1),\ (2,0).$$

They have degree six, rank three, group order sixty and tail order ten. The other primitive representatives either have no anchor with a transitive complementary tail or have an odd mark. We explicitly do not reuse the character condition from the earlier faithful-augmentation theorem: the present hypotheses supply no such character. If the exceptional list does not apply, $k < d/2$ implies (1) by integrality. This completes the inventory. □

Theorem 3. *Let an internal perfect split in a transitive natural symmetric C-marking of degree n and rank r, with $2r > n + 2$, be*

$$q = \rho_i = (a\ b)\beta, \qquad L = \langle \rho_0, \dots, \rho_{i-1} \rangle, \quad R = \langle \rho_{i+1}, \dots, \rho_{r-1} \rangle,$$

where L is transitive and supported on A, R is transitive and supported on B, and $A \sqcup B$ is the domain. Suppose $J = \langle q, R \rangle$ contains only even permutations. Then $L = \mathrm{Sym}(A)$ and either

$$2i > |A| + 2, \tag{2}$$

or $|B| = 5$, R has rank two and order ten, $|A|$ is even, and $i = |A|/2 + 1$.

Proof. The right wing is even and nontrivial, so is not its full symmetric group. The acquired actual-wing theorem therefore gives $L = \mathrm{Sym}(A)$. The group J is supported faithfully on $Y = \{a\} \cup B$. It is transitive there, and its stabilizer at a contains R, transitive on the complement. Thus J is doubly transitive, hence primitive. Its actual submarking inherits IP. Its first mark q moves a, and its tail fixes a.

The adjacent left mark ρ_{i-1} moves a: all earlier left marks fix a by commutation with the unique crossing, whereas L is transitive. Write $\rho_{i-1} = (a\ c)\gamma$. Its support is in A, so $c \in A \setminus \{a\}$ and γ has support disjoint from $Y \cup \{c\}$. This is exactly the actual outside attachment required in Theorem 2; no restriction kernel is being silently discarded.

Put $t = \mathrm{rank}(R)$ and $b = |B|$. In the ordinary case that theorem gives $2(t+1) \leq b$. Consequently

$$2i = 2r - 2 - 2t > |A| + b - 2t \geq |A| + 2,$$

which is (2). Otherwise $b = 5$ and $t = 2$. Then $2i > |A| + 1$; if (2) fails, the even integer $2i$ equals $|A| + 2$, giving the stated critical case and parity of $|A|$. □

Scope and attribution. The alternating and transitive classification theorems, their exact finite lists and their complete validation remain attributed to Cameron, Fernandes, Leemans and Mixer through L044 and L047. The A11 obstruction and central-intersection mechanism are acquired from earlier separately reviewed L019 components. The new finite work filters the complete A10 list and proves the changed even attachment interface, with its precise residual degree-six case. The intended high-rank extension problem is that of Cameron–Fernandes–Leemans, arXiv:2212.12723v2.

This is a descent reduction, not an extension theorem for the remaining critical case. In particular, a critical symmetric wing need not admit a split with two odd attaching groups: an explicit rank-seven S_{12} marking can have only one odd mark, which is pure. That diagnostic refutes a proposed strengthening of our own, not a source conclusion. Valid leaf kernels, the full reversible correspondence and the whole L019 principal claim remain pending. No new census, open conjecture, Lean theorem or compression measurement is claimed here.